Appendix II. Update to the Low-Income Country Debt Sustainability Analysis

This appendix provides an update to the joint IMF-World Bank LIC-DSA for the 2nd review under the concurrent ECF arrangement (IMF Country Report No. 13/192), in the context of the Guinean authorities' request for a disbursement under the Rapid Credit Facility to help meet a financing shortfall as a result of the Ebola outbreak. It includes updated projections for the macroeconomic framework and the Simandou mining and related infrastructure projects, and new information on borrowing. Guinea's risk of debt distress remains moderate.

1. Guinea continues to be assessed at a moderate risk of debt distress.[1] [2] This debt sustainability analysis (DSA) updates the LIC-DSA presented to the Board in May 2013, and uses a higher and unified discount rate of 5 percent (previously 3 percent) which was introduced in October 2013. A full update of the LIC-DSA will be prepared in the context of the next review (5th) under the ECF arrangement and Article IV consultation. As in the previous assessment, the moderate rating is primarily due to Guinea's vulnerability to adverse shocks to growth, exports, foreign direct investment flows, fiscal performance, and delays in mining projects coming on stream. The LIC-DSA underscores the importance of implementing a prudent debt policy and relying primarily on concessional loans and grant and avoid an excessive ramping up of external debt, especially in the short term, until the projected increase in economic growth on the back of mining sector output and of exports comes to fruition. In addition, the authorities' ongoing efforts to maintain macroeconomic stability, improve project implementation capacity and the business climate, establish a sound governance structure in the mining sector, and strengthen debt management should be continued. In addition, structural reforms aimed at removing bottlenecks, improving productivity, and supporting economic diversification are important to minimize the adverse effects of Dutch disease that could arise from the significant expansion of the mining sector.

2. Following the attainment of the HIPC initiative completion point, at end-2012 Guinea's external public debt amounted to $1.3 billion (23 percent of GDP). Since then, the accumulation of new debt (disbursed and outstanding) has been modest and at end-2013 external public debt had increased to $1.6 billion (25.3 percent of GDP). The increase is primarily accounted for by disbursements for the construction of the Kaleta hydroelectric dam. Guinea also has outstanding external debt arrears of $186 million (11 percent of total debt) which are virtually all to non-Paris club official bilateral creditors. These arrears have increased slightly since end-2012 because of arrears accrued on debt service falling due to a creditor, and discussions between the creditor and authorities are ongoing so that these can be resolved. The LIC-DSA, in line with the authorities objectives, incorporates a phased clearance of arrears through partial cancellation and rescheduling (in the context of the HIPC completion point) which is different from that assumed in the last LIC-DSA, reflecting progress thus far and expected.[3]

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[1] This Debt Sustainability Analysis (DSA) is based on the Debt Sustainability Framework (DSF) for low-income countries (LICs), approved by the Executive Boards of the IMF and World Bank.

[2] Guinea's three-year average CPIA score is 2.93 and its quality of policy and institutions rating is weak.

[3] The status of these arrears under the IMF's lending into arrears (LIA) and arrears to official creditors has not changed since the (last) 4th review under the ECF arrangement

3. There are several significant changes in the baseline macroeconomic assumptions compared to the last LIC DSA (Table 1). As regards the mining sector, a new model based estimation of the financial flows, in particular foreign direct investment, mining tax revenues, mining imports and exports, associated with two major mining projects (the Simandou iron ore mine and an expansion of the capacity of an existing bauxite mine) have been incorporated. In addition, a delay in the onset of construction of the Simandou mining and related infrastructure project has pushed out a sharp rise in the related FDI inflows and imports, as well as the onset of production and exports. The delay in the Simandou project stemmed from protracted negotiations on an overarching investment framework which were concluded successfully in mid-2014. Production and exports are now expected to begin in late 2019 and 2020 compared to 2017 in the last LIC-DSA; also the projected volume of iron ore exports is somewhat higher. The expansion of the bauxite mining project is now included as the timing and details have become available only recently; the scale of the expansion is modest and would result in a two-step increase in bauxite exports in 2017 and 2022. The LIC-DSA incorporates the economic consequences of the Ebola outbreak. While there are uncertainties about the duration and spread of the epidemic, at this point in time according to the World Health Organization (WHO) it may take 6-9 months before the outbreak is under control provided strong international action is taken immediately. On this basis, the epidemic is projected to have only a short-term temporary impact, with GDP growth lower and the basic primary fiscal deficit higher in 2014–15. Medium and long term growth is projected to be higher than in the last DSA. This reflects the higher iron ore output from the Simandou mining project, the impact of the new bauxite mine expansion project, higher electricity supply (based on recent progress in raising supply capacity and future developments), and a stronger structural reform impact.

4. There are also changes in a number of the baseline assumptions concerning debt. One new projected nonconcessional loan has been incorporated.[4] The loan, which has yet to be signed, is for the renovation and extension of the electricity transmission line between the Kaleta hydro-electric dam and Conakry. The loan is long-term and is expected to have a grant element of slightly less than 30 percent and amount to about $80 million (4.8 percent of end-2013 external debt stock and 1.2 percent of projected 2014 GDP). As regards to the large China Eximbank loan ($335 million) for the construction of the Kaleta dam that was signed in early 2013 and incorporated in the last LIC-DSA, progress in construction, and disbursements to date and projected based on an updated disbursement schedule, are broadly as initially assumed.[5] The LIC-DSA also includes the disbursement under the Rapid Credit Facility that the Guinean authorities have requested. The government's emergency response plan to the Ebola outbreak is not fully financed as yet and the authorities are seeking additional donor assistance in the form of grants to cover this financing need. In line with this the LIC-DSA incorporates the sought after financing as grants. As compared to the last LIC-DSA, the rate of debt accumulation in the outer years of the projection period is somewhat higher than before, reflecting a lower primary fiscal balance than in the last LIC-DSA. This reflects a more stable ratio of current expenditure to GDP, following a lengthy period of gradual downward adjustment, to avoid too great a compression of current expenditures, while maintaining the investment effort. The net combination of a higher discount rate,

[4] At the time of the 4th review under the ECF arrangement, the performance criterion on new nonconcessional borrowing was adjusted to permit the loan under the ECF-supported program.

[5] The disbursement schedule assumed in the last LIC-DSA was $160 million in 2013, $102 million in 2014, $23 million in 2015, and $48 million in 2016. In the current LIC-DSA the disbursement schedule is $178 million in 2013, $98 million in 2014, $12 million in 2015, and $48 million in 2016.

higher external borrowing and slightly higher level of nominal GDP, results in slightly higher debt indicators in terms of GDP in the later years of the projection period. By contrast the PV of external debt is considerably lower in the near term reflecting the higher discount rate and a slightly lower initial external debt stock.[6]

5. The results of the external LIC-DSA confirm that Guinea's debt dynamics are sustainable. Under the baseline all debt indicators remain below the policy-dependent thresholds.[7] However, the stress tests and alternative scenarios show some breaches of the thresholds, and are indicative of Guinea's vulnerability to adverse shocks to growth, exports, foreign direct investment flows and fiscal performance. The inclusion of domestic debt does not significantly change the conclusions of the external LIC-DSA, as no major recourse to the domestic financial market is assumed given the shallow undeveloped nature of the this market. As regards to the planned recapitalization of the central bank by the government, following a first injection further planning recapitalization has been put on hold pending an evaluation of the central bank's capital need.

6. The authorities broadly agreed with the assumptions and conclusions of this update of the LIC-DSA.

[6] The outstanding external public debt stock at end-2013 is somewhat less than projected in the last LIC-DSA—$1.577 billion compared to $1.637 billion.

[7] In Table 3, the relatively large external debt dynamic residuals, after taking into account exceptional financing, reflect weaknesses in the coverage of balance of payments data.

Table 1. Guinea: Structure of External Public Debt

	end-2013 US Dollars (millions)	end-2013 Percent of Total	end-2013 Percent of GDP
Total	**1,577.3**	**100.0**	**25.3**
Multilateral Creditors	**588.3**	**37.3**	**9.4**
IMF	85.6	5.4	1.4
World Bank	168.1	10.7	2.7
AfDB Group	95.1	6.0	1.5
IsDB	122.2	7.7	2.0
EU	0.0	0.0	0.0
Other Multilateral Creditors	117.3	7.4	1.9
Official Bilateral Creditors	**923.5**	**58.5**	**14.8**
Paris Club	288.8	18.3	4.6
Non-Paris Club	498.0	31.6	8.0
Arab Funds	136.6	8.7	2.2
Commercial Creditors	**65.6**	**4.2**	**1.1**
Memo			
Arrears [1/]	186.1	...	3.0

Sources: Guinean authorities and IMF and World Bank estimates.

1/ The arrears are primarily to some non-Paris Club official bilateral creditors ($88.5 million) and commercial creditors ($65.6 million) and concern debt outstanding on which there is no remaining debt service obligations falling due. The Guinean authorities have invited these creditors for discussions on debt relief and a normalization of the arrears. In addition, there are some arrears to a Paris Club creditor for which a resolution is under discussion with the Guinean authorities.

Table 2. Guinea: LIC DSA Macroeconomic Assumptions, 2014–32

(Percent of GDP, unless otherwise indicated)

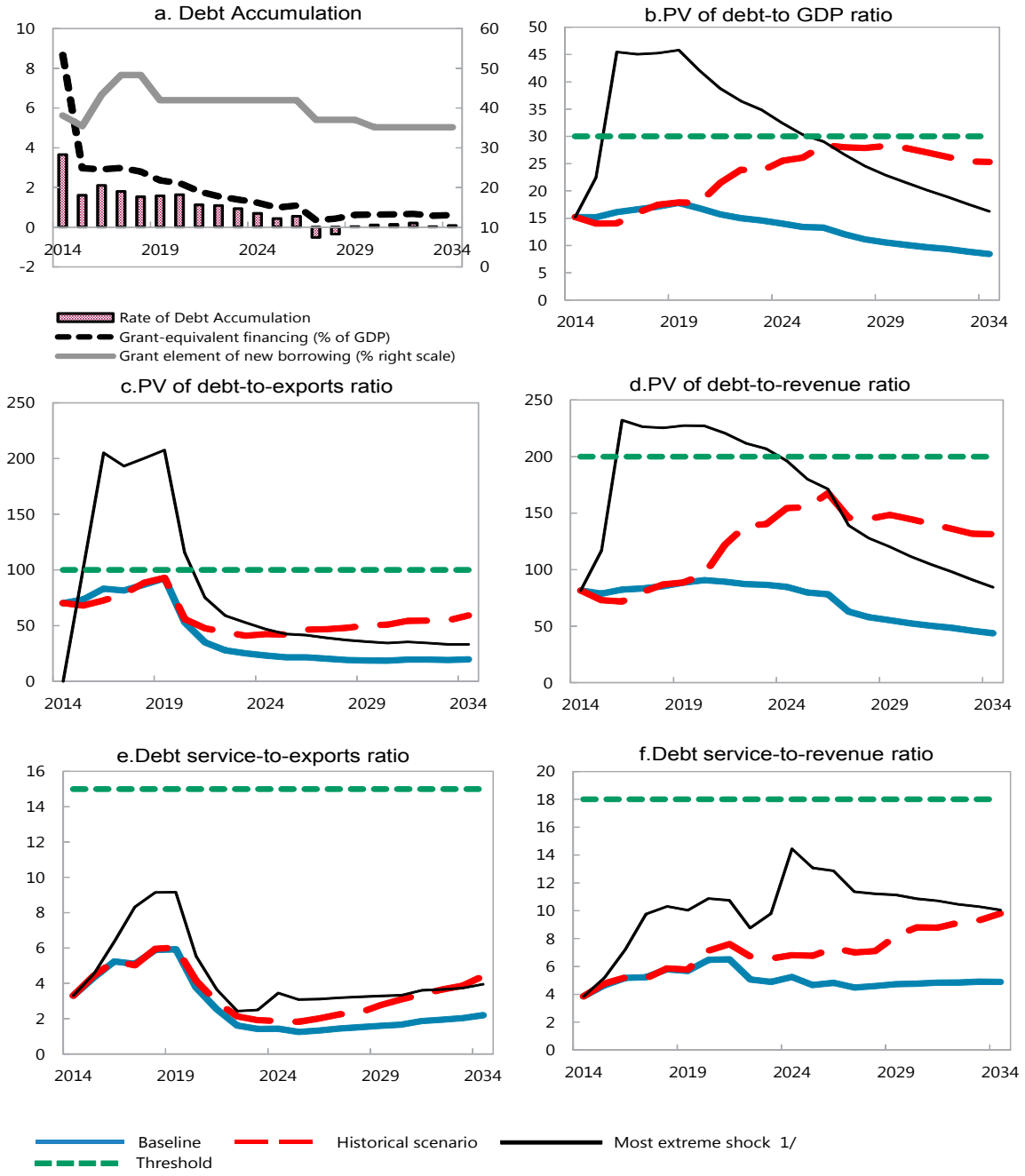
	Previous LIC DSA					Current LIC DSA				
	2014	2015	2020	2025	2032	2014	2015	2020	2025	2032
Nominal GDP ($ Million)	6,347.9	6,797.7	14,092.3	17,314.9	24,296.4	6,769.7	7,455.3	11,313.2	18,683.2	27,323.2
Real GDP (Percentage change)	5.2	5.4	6.4	3.2	3.6	2.4	4.1	7.4	5.0	4.4
Nominal GDP, excl. megaprojects ($ Million)	6,347.9	6,797.7	9,447.5	12,746.7	19,905.9	6,769.7	7,455.3	10,270.1	14,043.6	21,966.4
Real GDP, excl. megaprojects (Percentage change)	5.2	5.4	4.2	4.4	4.8	2.4	4.1	4.2	4.4	4.8
Fiscal Accounts										
Revenue and Grants	24.0	23.6	25.9	26.5	26.9	23.9	21.3	21.4	23.2	24.4
Primary Expenditure	25.5	24.1	26.6	26.0	26.1	28.7	22.9	22.8	23.1	24.8
Of which : Capital expenditure and net lending	10.8	9.4	13.0	13.9	17.4	12.6	7.3	9.7	10.2	12.2
Primary Fiscal Balance	-1.5	-0.6	-0.7	0.5	0.8	-4.8	-1.6	-1.4	0.0	-0.4
New external borrowing [2/]	3.1	1.7	1.6	0.5	0.1	5.6	2.6	3.5	1.3	1.3
Grant element of new external borrowing (%)	15.5	20.3	23.0	23.0	17.4	38.1	35.5	42.0	42.0	35.2
Balance of Payments										
Exports of goods and services	28.0	26.9	52.6	51.5	37.5	21.7	20.6	35.1	82.8	59.8
Imports of goods and services	-73.1	-67.8	-49.0	-46.8	-40.3	-39.9	-39.6	-52.1	-72.6	-55.8
Current account (including official transfers)	-46.9	-42.1	-7.0	-5.8	-10.5	-18.1	-18.0	-18.8	-11.5	-10.5
Foreign direct investment	41.8	35.8	3.6	3.9	4.6	5.6	10.1	12.1	7.0	4.6

Source: Guinean authorities, IMF and World Bank staff estimates.

[1/] The LIC-DSA and Figures 1-2 and Tables 3-6 use total GDP in the calculations and the ratios expressed in terms of GDP. The ratios to GDP in this Table 2 are expressed in terms of GDP excluding Simandou and are consistent with tables in the main text, figures, and tables in the Staff Report.

[2/] Includes publicly guaranteed external borrowing.

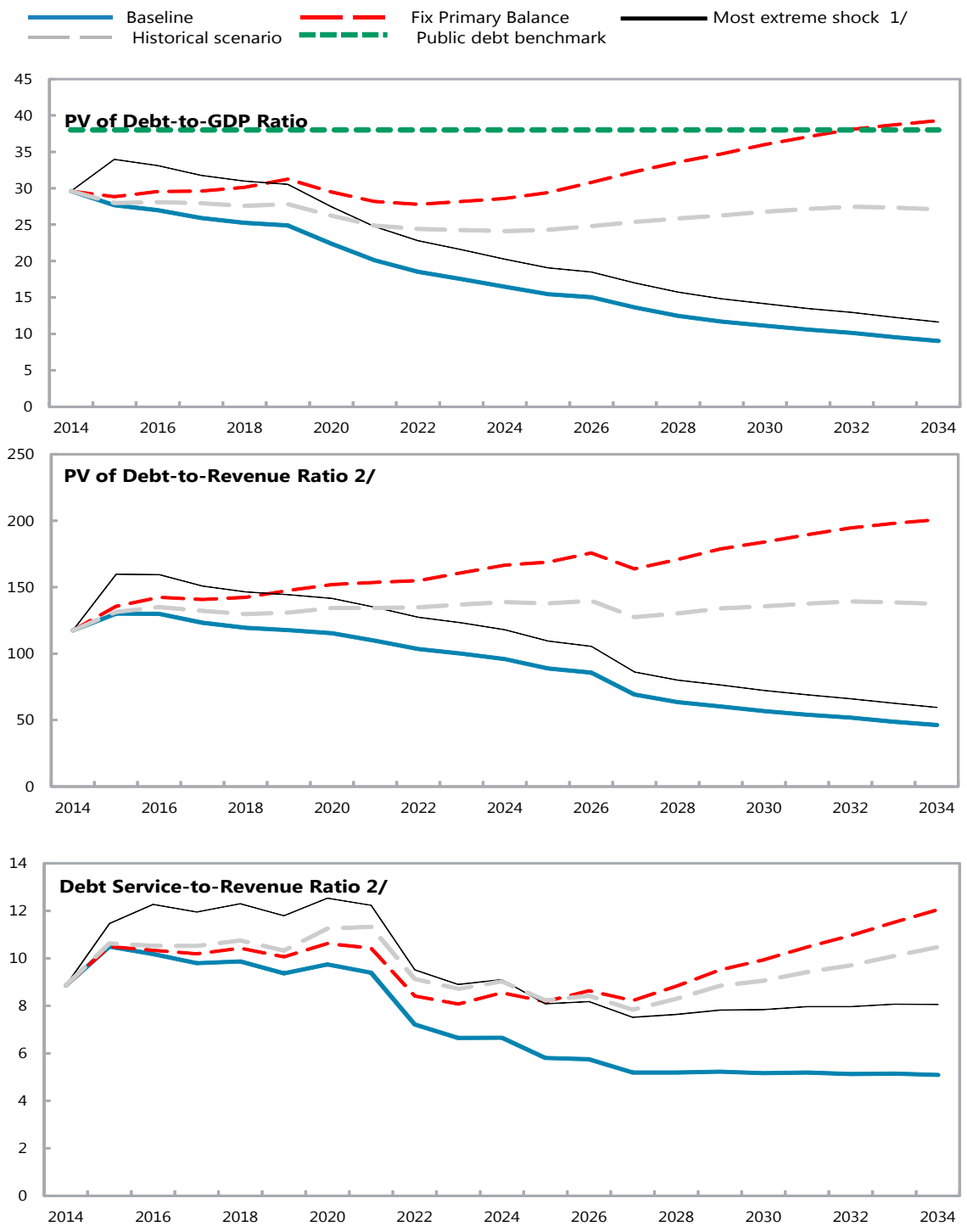
Figure 1. Guinea: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014–34[1/]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Combination shock and in figure f. to a Combination shock

Figure 2. Guinea: Indicators of Public Debt Under Alternative Scenarios, 2014–34[1/]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 3. Guinea: External Debt Sustainability Framework, Baseline Scenario, 2011–34[1/]

(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**67.5**	**23.0**	**25.7**			**29.1**	**28.1**	**29.6**	**30.5**	**31.7**	**32.6**		**24.6**	**14.5**	
of which: public and publicly guaranteed (PPG)	67.5	23.0	25.7			29.1	28.1	29.6	30.5	31.7	32.6		24.6	14.5	
Change in external debt	-3.2	-44.5	2.7			3.4	-0.9	1.5	1.0	1.1	0.9		-1.2	-0.6	
Identified net debt-creating flows	10.2	8.2	15.5			10.7	6.8	5.9	1.9	2.7	4.3		0.6	2.6	
Non-interest current account deficit	**18.4**	**25.2**	**21.0**	**10.5**	**8.6**	**16.6**	**17.8**	**33.9**	**52.4**	**52.7**	**57.4**		**7.5**	**6.3**	8.4
Deficit in balance of goods and services	19.1	24.2	18.7			18.2	19.0	35.0	52.6	52.4	55.9		-7.1	-3.8	
Exports	31.5	29.8	24.7			21.7	20.6	19.4	20.4	19.8	19.3		60.4	42.8	
Imports	50.6	54.1	43.4			39.9	39.6	54.3	72.9	72.1	75.1		53.3	39.1	
Net current transfers (negative = inflow)	-8.5	-6.2	-4.3	-6.5	1.7	-7.2	-5.2	-4.8	-3.9	-3.8	-3.6		-2.2	-1.4	-2.0
of which: official	-2.2	-1.0	-0.4			-2.8	-1.1	-0.3	-0.4	-0.3	-0.3		-0.2	-0.2	
Other current account flows (negative = net inflow)	7.8	7.1	6.6			5.6	4.0	3.6	3.8	4.1	5.1		16.8	11.6	
Net FDI (negative = inflow)	**-6.5**	**-11.7**	**-3.4**	**-5.3**	**3.1**	**-5.6**	**-10.1**	**-27.0**	**-49.1**	**-49.2**	**-52.2**		**-5.9**	**-3.3**	-5.4
Endogenous debt dynamics 2/	**-1.7**	**-5.4**	**-2.0**			**-0.4**	**-0.8**	**-1.0**	**-1.4**	**-0.8**	**-0.9**		**-0.9**	**-0.4**	
Contribution from nominal interest rate	1.0	0.7	0.2			0.2	0.3	0.3	0.4	0.4	0.4		0.3	0.2	
Contribution from real GDP growth	-2.7	-2.3	-0.5			-0.6	-1.1	-1.3	-1.8	-1.2	-1.3		-1.3	-0.6	
Contribution from price and exchange rate changes	0.0	-3.7	-1.7			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-13.4**	**-52.7**	**-12.8**			**-7.3**	**-7.7**	**-4.5**	**-0.9**	**-1.6**	**-3.4**		**-1.8**	**-3.2**	
of which: exceptional financing	-1.6	-39.9	0.0			-1.9	-0.4	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	12.7			15.2	15.2	16.1	16.6	17.1	17.8		14.0	8.4	
In percent of exports	...	...	51.3			70.2	73.5	83.2	81.5	86.8	92.5		23.2	19.7	
PV of PPG external debt	**...**	**...**	**12.7**			**15.2**	**15.2**	**16.1**	**16.6**	**17.1**	**17.8**		**14.0**	**8.4**	
In percent of exports	**...**	**...**	**51.3**			**70.2**	**73.5**	**83.2**	**81.5**	**86.8**	**92.5**		**23.2**	**19.7**	
In percent of government revenues	**...**	**...**	**69.1**			**81.6**	**78.9**	**82.3**	**83.4**	**85.4**	**88.5**		**84.8**	**43.8**	
Debt service-to-exports ratio (in percent)	**13.7**	**15.2**	**3.9**			**3.3**	**4.3**	**5.2**	**5.1**	**5.9**	**5.9**		**1.4**	**2.2**	
PPG debt service-to-exports ratio (in percent)	**13.7**	**15.2**	**3.9**			**3.3**	**4.3**	**5.2**	**5.1**	**5.9**	**5.9**		**1.4**	**2.2**	
PPG debt service-to-revenue ratio (in percent)	**25.8**	**22.5**	**5.2**			**3.9**	**4.6**	**5.2**	**5.2**	**5.8**	**5.7**		**5.2**	**4.9**	
Total gross financing need (Billions of U.S. dollars)	0.8	1.0	1.2			0.8	0.6	0.6	0.4	0.4	0.6		0.4	1.2	
Non-interest current account deficit that stabilizes debt ratio	21.6	69.7	18.3			13.2	18.7	32.4	51.4	51.6	56.5		8.7	7.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.9	3.8	2.3	2.6	1.4	2.4	4.1	5.0	6.5	4.1	4.2	4.4	5.4	4.5	5.1
	0.0	5.9	8.1	4.4	15.4	6.1	5.8	3.4	1.5	1.8	1.0	3.3	3.5	1.4	2.7
Effective interest rate (percent) 5/	1.4	1.2	0.8	1.4	0.4	1.0	1.0	1.2	1.3	1.3	1.3	1.2	1.4	1.5	1.4
Growth of exports of G&S (US dollar terms, in percent)	12.3	4.3	-8.3	6.4	13.6	-4.7	4.7	2.0	13.8	2.7	2.7	3.5	14.1	-2.1	16.2
Growth of imports of G&S (US dollar terms, in percent)	48.8	17.6	-11.2	13.6	19.3	-0.1	9.3	49.0	45.1	4.8	9.6	19.6	7.3	-4.7	3.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	38.1	35.5	43.4	48.4	48.4	42.0	42.6	42.0	35.2	38.7
Government revenues (excluding grants, in percent of GDP)	16.8	20.1	18.4			18.7	19.2	19.6	19.9	20.1	20.1		16.5	19.3	18.3
Aid flows (in Billions of US dollars) 7/	0.2	0.2	0.1			0.6	0.2	0.3	0.4	0.4	0.3		0.3	0.3	
of which: Grants	0.2	0.2	0.1			0.4	0.2	0.1	0.1	0.1	0.1		0.1	0.1	
of which: Concessional loans	0.0	0.0	0.0			0.1	0.1	0.2	0.3	0.3	0.2		0.2	0.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			8.7	3.0	2.9	3.0	2.8	2.4		1.2	0.6	1.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			71.4	64.0	56.4	60.2	60.3	56.1		59.3	50.1	77.1
Memorandum items:															
Nominal GDP (Billions of US dollars)	5.1	5.6	6.2			6.8	7.5	8.1	8.8	9.3	9.8		17.3	30.7	
Nominal dollar GDP growth	4.0	9.9	10.6			8.7	10.1	8.6	8.1	6.0	5.2	7.8	9.0	6.0	8.0
PV of PPG external debt (in Billions of US dollars)			0.8			1.0	1.1	1.3	1.4	1.6	1.7		2.4	2.5	
(PVt-PVt-1)/GDPt-1 (in percent)						3.7	1.6	2.1	1.8	1.5	1.6	2.0	0.7	0.1	0.4
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	12.7			15.2	15.2	16.1	16.6	17.1	17.8		14.0	8.4	
PV of PPG external debt (in percent of exports + remittances)	...	...	51.3			70.2	73.5	83.2	81.5	86.8	92.5		23.2	19.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.9			3.3	4.3	5.2	5.1	5.9	5.9		1.4	2.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(Percent)

				Projections				
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	15	15	16	17	17	18	**14**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	15	14	14	16	17	18	**26**	25
A2. New public sector loans on less favorable terms in 2014-2034 2	15	16	17	19	20	22	**19**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	15	15	17	17	18	19	**15**	9
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	15	16	20	20	20	21	**16**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	15	18	22	22	23	24	**19**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	15	20	37	36	37	37	**26**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	15	22	45	45	45	46	**32**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	15	21	22	23	23	24	**19**	11
PV of debt-to-exports ratio								
Baseline	70	74	83	81	87	92	**23**	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	70	68	73	78	88	93	**42**	59
A2. New public sector loans on less favorable terms in 2014-2034 2	70	75	89	92	103	113	**32**	35
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	70	73	81	79	85	90	**23**	19
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	70	90	125	121	128	135	**33**	26
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	70	73	81	79	85	90	**23**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	70	99	189	178	185	192	**43**	31
B5. Combination of B1-B4 using one-half standard deviation shocks		102	205	193	200	208	**47**	33
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	70	73	81	79	85	90	**23**	19
PV of debt-to-revenue ratio								
Baseline	82	79	82	83	85	89	**85**	44
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	82	73	72	80	87	89	**154**	131
A2. New public sector loans on less favorable terms in 2014-2034 2	82	81	88	94	101	108	**117**	78
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	82	80	86	87	89	92	**88**	45
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	82	86	100	100	101	104	**96**	47
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	82	93	111	112	115	119	**114**	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	82	106	187	183	182	184	**158**	68
B5. Combination of B1-B4 using one-half standard deviation shocks	82	117	232	226	225	227	**196**	84
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	82	109	112	114	116	120	**115**	59

Table 4. Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	3	4	5	5	6	6	**1**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	3	4	5	5	6	6	**2**	4
A2. New public sector loans on less favorable terms in 2014-2034 2	3	4	5	5	6	7	**2**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	3	4	5	5	6	6	**1**	2
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	3	5	7	7	8	8	**2**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	3	4	5	5	6	6	**1**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	3	4	6	8	8	8	**3**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	3	5	6	8	9	9	**3**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	3	4	5	5	6	6	**1**	2
Debt service-to-revenue ratio								
Baseline	4	5	5	5	6	6	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	5	5	5	6	6	**7**	10
A2. New public sector loans on less favorable terms in 2014-2034 2	4	5	5	5	6	7	**6**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	5	6	6	6	6	**6**	5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	5	5	6	6	6	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	6	7	7	8	8	**7**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	5	6	8	8	8	**12**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	7	10	10	10	**14**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	6	7	7	8	8	**7**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	36	36	36	36	36	36	**36**	36

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 5. Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	77.8	35.4	39.5			43.4	40.6	40.4	39.8	39.8	39.6		27.0	15.1	
of which: foreign-currency denominated	67.5	23.0	25.7			29.1	28.1	29.6	30.5	31.7	32.6		24.6	14.5	
Change in public sector debt	-21.9	-42.3	4.0			4.0	-2.8	-0.2	-0.6	-0.1	-0.1		-1.7	-0.7	
Identified debt-creating flows	-10.4	-42.4	2.5			2.8	-1.6	-0.9	-0.9	-0.2	-0.4		-1.7	-0.1	
Primary deficit	-0.7	1.6	4.1	1.5	4.7	4.8	1.6	1.2	1.5	1.5	1.0	1.9	0.3	0.6	0.1
Revenue and grants	20.2	22.9	19.8			23.9	21.3	20.8	21.0	21.2	21.2		17.2	19.6	
of which: grants	3.4	2.7	1.5			5.2	2.1	1.2	1.1	1.1	1.0		0.6	0.3	
Primary (noninterest) expenditure	19.5	24.4	23.9			28.7	22.9	22.0	22.5	22.6	22.2		17.5	20.2	
Automatic debt dynamics	-8.1	-10.8	-1.6			-2.0	-3.2	-2.1	-2.4	-1.6	-1.4		-2.0	-0.7	
Contribution from interest rate/growth differential	-7.7	-3.4	-0.7			-1.1	-1.9	-2.0	-2.5	-1.7	-1.7		-1.7	-0.8	
of which: contribution from average real interest rate	-4.0	-0.5	0.1			-0.2	-0.2	-0.1	-0.1	-0.2	-0.1		-0.2	-0.1	
of which: contribution from real GDP growth	-3.7	-2.9	-0.8			-0.9	-1.7	-1.9	-2.5	-1.6	-1.6		-1.5	-0.7	
Contribution from real exchange rate depreciation	-0.4	-7.4	-0.9			-0.8	-1.3	-0.1	0.1	0.1	0.3		...	...	
Other identified debt-creating flows	-1.6	-33.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-1.6	-33.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-11.4	0.0	1.5			1.2	-1.2	0.8	0.3	0.1	0.2		0.0	-0.7	
Other Sustainability Indicators															
PV of public sector debt	...	...	26.5			29.6	27.7	27.0	25.9	25.2	24.9		16.5	9.0	
of which: foreign-currency denominated	...	...	12.7			15.2	15.2	16.1	16.6	17.1	17.8		14.0	8.4	
of which: external	...	...	12.7			15.2	15.2	16.1	16.6	17.1	17.8		14.0	8.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.7	7.1	6.0			7.0	3.9	3.3	3.6	3.6	3.0		1.5	1.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	133.4			123.8	130.0	129.9	123.1	119.4	117.6		95.9	46.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	144.2			158.6	143.9	137.8	130.2	125.7	123.6		99.5	46.9	
of which: external 3/	...	...	69.1			81.6	78.9	82.3	83.4	85.4	88.5		84.8	43.8	
Debt service-to-revenue and grants ratio (in percent) 4/	26.9	24.3	9.8			9.3	10.5	10.2	9.8	9.9	9.4		6.7	5.1	
Debt service-to-revenue ratio (in percent) 4/	32.4	27.6	10.6			12.0	11.6	10.8	10.4	10.4	9.8		6.9	5.2	
Primary deficit that stabilizes the debt-to-GDP ratio	21.2	43.9	0.0			0.8	4.4	1.4	2.1	1.6	1.1		2.0	1.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.9	3.8	2.3	2.6	1.4	2.4	4.1	5.0	6.5	4.1	4.2	4.4	5.4	4.5	5.1
Average nominal interest rate on forex debt (in percent)	1.4	1.2	0.8	1.4	0.3	1.0	1.0	1.2	1.3	1.3	1.3	1.2	1.4	1.5	1.4
Average real interest rate on domestic debt (in percent)	-12.5	-1.2	2.1	-5.5	6.7	-0.1	0.3	0.5	1.2	0.9	1.7	0.7	-2.5	-0.4	-1.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.5	-11.5	-4.1	0.6	19.7	-3.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	19.7	13.0	6.3	17.5	9.5	7.4	5.8	7.3	6.6	6.9	6.1	6.7	8.6	6.4	7.8
Growth of real primary spending (deflated by GDP deflator, in percent)	-26.7	29.9	0.2	0.4	13.4	22.9	-16.9	0.7	9.4	4.4	2.1	3.8	2.5	5.5	4.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	38.1	35.5	43.4	48.4	48.4	42.0	42.6	42.0	35.2	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 6. Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2014–34

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	30	28	27	26	25	25	16	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	30	28	28	28	28	28	24	27
A2. Primary balance is unchanged from 2014	30	29	30	30	30	31	29	39
A3. Permanently lower GDP growth 1/	30	28	27	26	26	26	19	16
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	30	29	30	30	30	30	23	21
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	30	31	33	32	31	30	20	12
B3. Combination of B1-B2 using one half standard deviation shocks	30	30	31	31	31	31	23	19
B4. One-time 30 percent real depreciation in 2015	30	33	32	30	28	28	18	12
B5. 10 percent of GDP increase in other debt-creating flows in 2015	30	34	33	32	31	31	20	12
PV of Debt-to-Revenue Ratio 2/								
Baseline	117	130	130	123	119	118	96	46
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	117	131	135	132	130	131	139	137
A2. Primary balance is unchanged from 2014	117	136	142	141	142	148	167	201
A3. Permanently lower GDP growth 1/	117	131	131	125	122	122	108	82
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	117	135	144	140	140	142	135	108
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	117	144	159	150	146	144	118	60
B3. Combination of B1-B2 using one half standard deviation shocks	117	139	151	146	145	146	134	99
	117	157	153	141	135	131	107	62
B5. 10 percent of GDP increase in other debt-creating flows in 2015	117	160	159	151	146	144	118	59
Debt Service-to-Revenue Ratio 2/								
Baseline	9	10	10	10	10	9	7	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	11	11	11	11	10	9	10
A2. Primary balance is unchanged from 2014	9	10	10	10	10	10	9	12
A3. Permanently lower GDP growth 1/	9	11	10	10	10	10	7	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	9	11	11	11	11	10	8	8
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	9	10	11	11	11	10	8	6
B3. Combination of B1-B2 using one half standard deviation shocks	9	11	11	11	11	10	8	8
0	9	11	12	12	12	12	9	8
B5. 10 percent of GDP increase in other debt-creating flows in 2015	9	10	11	11	11	10	8	6

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.